

# Consistency Models in Distributed Systems

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# Overview

- A **consistency model** defines the rules that determine when updates made to data become visible to other users or processes in a distributed system.
- When data is replicated across multiple servers, different nodes may temporarily contain different values. Consistency models specify what guarantees the system provides.

# 1. Strong Consistency (Linearizability)

- **Definition:** Once a **write operation completes**, every **subsequent read returns the latest value**.
- **Example**
  - Suppose a bank account balance is ₹10,000.
  - User A withdraws ₹1,000.
  - New balance becomes ₹9,000.
  - Any user reading the balance immediately sees ₹9,000.
- **Advantages**
  - Easy to understand.
  - No stale data.
- **Disadvantages**
  - Higher latency.
  - Requires coordination among replicas.

## Applications

- Banking systems
- Payment systems
- Stock trading

## 2. Sequential Consistency

**Definition:** All **processes see operations in the same order**, but that order may not match real-time order.

- **Example**

- Two writes occur:
  - W1:  $x = 10$
  - W2:  $x = 20$

- All users must see: 10->20

- **Applications**

- Shared memory systems
- Distributed computing

# 3. Causal Consistency

**Definition:** Operations that are **causally related must be observed in the same order by everyone.**

- **Example**

- On a social network:
  - Shyam posts:
    - "Exam is tomorrow."
  - Ram replies:
    - "Thanks for the reminder."
- Everyone must see:
  - Shyam's post
    - ↓
  - Ram's reply
- Seeing the reply before the original post would violate causal consistency.

## Applications

- Social media
- Collaborative editing
- Messaging systems

# 4. Eventual Consistency

- **Definition:** If no new updates occur, all replicas eventually converge to the same value.

- **Example**

- A product quantity changes:
- Inventory = 50  $\rightarrow$  45
- Replica A updates immediately.
- A = 45
- B = 50
- C = 50
- For a short period, users may see different values.

- **After Synchronization**

- A = B = C = 45

## Applications

- DNS systems
- Large NoSQL databases
- Social media feeds

# 5. Weak Consistency

- **Definition:** No guarantee that reads immediately reflect recent writes.
- **Example**
  - A cached weather service:
  - Actual Temperature = 30°C
  - Cache still shows = 28°C
- Users may receive outdated information.

## Applications

- Caching systems
- Monitoring dashboards

## 6. Read-Your-Writes Consistency

- **Definition:** After a user performs a write, that same user always sees the updated value.
- **Example**
  - You update your profile name:
  - Name: Rahul → Rahul Kumar
  - Immediately after saving:
  - Your read → Rahul Kumar
- Other users may still see the old name temporarily.

### Applications

- User profiles
- Account settings

# 7. Monotonic Read Consistency

**Definition:** Once a process has seen a value, it will never see an older value later.

- **Example**

- A user reads:
    - Version 5
- Later reads:
  - Version 6 ✓
  - Version 7 ✓
  - Version 4 ✗
- The system prevents moving backward in time.

## Applications

- News feeds
- Email systems

# 8. Monotonic Write Consistency

- **Definition:** Writes from the same client are executed in the order they were issued.
- **Example**
  - A user performs:
    - Write 1: Address = Delhi
    - Write 2: Address = Mumbai
- All replicas must process:
  - Delhi → Mumbai
- Never:
  - Mumbai → Delhi

## Applications

- User profile updates
- Configuration management

# 9. Session Consistency

- **Definition:** Guarantees consistency within a user's session.

Usually includes:

- Read-your-writes
- Monotonic reads
- Monotonic writes
- **Example**
- During one login session:
  - Update password  
↓  
Read password settings  
↓  
See latest password
- The user always sees their own updates.

## Applications

- Web applications
- E-commerce websites

# Comparison

| Model                    | Latest Data Guaranteed? | Availability | Performance |
|--------------------------|-------------------------|--------------|-------------|
| Strong<br>(Linearizable) | Yes                     | Lower        | Lower       |
| Sequential               | Ordered                 | Medium       | Medium      |
| Causal                   | Causal order only       | Good         | Good        |
| Session                  | Within session          | Good         | Good        |
| Eventual                 | No                      | Very High    | Very High   |
| Weak                     | No                      | Highest      | Highest     |

Thank You